

## DBSE AND DBD

### MYSQL Triggers:

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Microsoft Windows [Version 10.0.26200.9278]
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C:\Users\Yagnesh katukala>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE trigger_demo;
Query OK, 1 row affected (0.08 sec)

mysql> USE trigger_demo;
Database changed
mysql> CREATE TABLE employees(
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
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mysql> USE trigger_demo;
Database changed
mysql> CREATE TABLE employees(
  ->     emp_id INT PRIMARY KEY,
  ->     emp_name VARCHAR(50),
  ->     dept_id INT,
  ->     salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql> DESC employees;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| emp_id | int           | NO   | PRI | NULL    |       |
| emp_name | varchar(50)   | YES  |     | NULL    |       |
| dept_id | int           | YES  |     | NULL    |       |
| salary  | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> INSERT INTO employees VALUES
  -> (101, 'Rahul', 1, 50000),
  -> (102, 'Priya', 2, 45000),
  -> (103, 'Kiran', 1, 55000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM employees;

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mysql> SELECT * FROM employees;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary |
+-----+-----+-----+-----+
| 101    | Rahul    | 1       | 50000.00 |
| 102    | Priya    | 2       | 45000.00 |
| 103    | Kiran    | 1       | 55000.00 |
+-----+-----+-----+-----+
3 rows in set (0.02 sec)

mysql> CREATE TABLE employee_audit(
  ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
  ->     emp_id INT,
  ->     emp_name VARCHAR(50),
  ->     action_type VARCHAR(20),
  ->     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER $$
mysql> CREATE TRIGGER after_employee_insert
  -> AFTER INSERT ON employees
  -> FOR EACH ROW
  -> BEGIN
  ->     INSERT INTO employee_audit
  ->     (emp_id, emp_name, action_type)
  ->     VALUES
  ->     (NEW.emp_id, NEW.emp_name, 'INSERT');
  -> END $$
Query OK, 0 rows affected (0.02 sec)

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mysql> DELIMITER ;
mysql>
mysql> INSERT INTO employees
  -> VALUES (106, 'Arjun', 1, 52000);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM employee_audit;
+-----+-----+-----+-----+-----+
| audit_id | emp_id | emp_name | action_type | action_time |
+-----+-----+-----+-----+-----+
| 1 | 106 | Arjun | INSERT | 2026-09-02 14:45:17 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DELETE FROM employees
  -> WHERE emp_id = 106;
Query OK, 1 row affected (0.03 sec)

mysql> We can automatically record that deletion.
  -> DELIMITER $$
  -> CREATE TRIGGER after_employee_delete
  -> AFTER DELETE ON employees
  -> FOR EACH ROW
  -> BEGIN
  -> INSERT INTO employee_audit
  -> (emp_id, emp_name, action_type)
  -> VALUES
  -> (OLD.emp_id, OLD.emp_name, 'DELETE');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server vers
for the right syntax to use near 'We can automatically record that deletion.'

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DELIMITER' at line 1
mysql> DELIMITER $$
mysql> CREATE TRIGGER after_employee_delete
  -> AFTER DELETE ON employees
  -> FOR EACH ROW
  -> BEGIN
  -> INSERT INTO employee_audit
  -> (emp_id, emp_name, action_type)
  -> VALUES
  -> (OLD.emp_id, OLD.emp_name, 'DELETE');
  -> END $$
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> UPDATE employees
  -> SET salary = 60000
  -> WHERE emp_id = 101;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELIMITER $$
mysql> CREATE TRIGGER after_employee_update
  -> AFTER UPDATE ON employees
  -> FOR EACH ROW
  -> BEGIN
  -> INSERT INTO employee_audit
  -> (emp_id, emp_name, action_type)
  -> VALUES
  -> (NEW.emp_id, NEW.emp_name, 'UPDATE');
  -> END $$

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mysql> DELIMITER ;
mysql>
mysql> INSERT INTO employees
-> VALUES (107, 'Ravi', 1, 15000);
ERROR 1644 (45000): Salary cannot be less than 20000
mysql> INSERT INTO employees VALUES (108, 'Meena', 1, 30000);
Query OK, 1 row affected (0.02 sec)

mysql> DELIMITER $$
mysql> CREATE TRIGGER prevent_salary_reduction
-> BEFORE UPDATE ON employees
-> FOR EACH ROW
-> BEGIN
->     IF NEW.salary < OLD.salary THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Salary reduction is not allowed';
->     END IF;
-> END $$
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> SELECT * FROM employee_audit;
+-----+-----+-----+-----+-----+
| audit_id | emp_id | emp_name | action_type | action_time |
+-----+-----+-----+-----+-----+
| 1 | 106 | Arjun | INSERT | 2026-09-02 14:45:17 |
| 2 | 108 | Meena | INSERT | 2026-09-02 14:48:19 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

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Search

